

## GREIFSWALDER OIE, Germany

WMO: 100970

Lat: 54.2436N

Lon: 13.9103E

Elev: 37

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 01-13

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 2             | 19.7       | 22.8 | 13.4                          | 10.9 | 21.8 | 16.7 | 12.8 | 24.6 | 42.5                  | 40.2 | 39.0 | 37.7 | 13.6                  | 240  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.7                    | 75.0            | 67.2 | 72.9 | 66.1 | 71.1 | 65.0 | 68.6                | 73.3 | 67.3 | 71.6 | 65.9 | 69.9 | 10.2                 | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.7                            | 98.4 | 71.7 | 65.3 | 93.7 | 70.4 | 64.0 | 89.2 | 69.1 | 32.9          | 73.6 | 31.7 | 71.7 | 30.7 | 70.0 | 73.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |     |            |     |            |     |      |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-----|------------|-----|------------|-----|------|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |     | n=20 years |     | n=50 years |     |      |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max | Min        | Max | Min        | Max |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l) | (m)        | (n) | (o)        | (p) |      |  |
| (4) 37.5          | 33.7 | 30.2 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A | N/A        | N/A | N/A        | N/A |      |  |
| (5)               |      |      | DB                         | 14.9 | 69.9               | 5.1 | 2.2                                                | 11.3 | 71.4       | 8.4 | 72.7       | 5.5 | 73.9       | 1.8 | 75.5 |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |     |            |     |            |     |      |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 48.8        | 33.9  | 33.3  | 37.4  | 44.7  | 53.6  | 60.3  | 65.9  | 65.7  | 59.6  | 51.4  | 42.9  | 35.4  |
|                                                                        | DBStd       | 12.76       | 6.21  | 5.15  | 5.12  | 4.64  | 4.59  | 3.79  | 4.16  | 3.52  | 3.87  | 4.86  | 4.39  | 6.23  |
|                                                                        | HDD50       | 2251        | 499   | 468   | 390   | 167   | 19    | 0     | 0     | 0     | 0     | 43    | 213   | 452   |
|                                                                        | HDD65       | 6058        | 964   | 888   | 855   | 608   | 353   | 148   | 38    | 34    | 168   | 423   | 662   | 917   |
|                                                                        | CDD50       | 1805        | 0     | 0     | 0     | 9     | 132   | 309   | 494   | 488   | 287   | 85    | 1     | 0     |
|                                                                        | CDD65       | 135         | 0     | 0     | 0     | 0     | 0     | 7     | 67    | 57    | 4     | 0     | 0     | 0     |
|                                                                        | CDH74       | 112         | 0     | 0     | 0     | 0     | 1     | 2     | 86    | 23    | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 4           | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 1     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 15.5        | 18.1  | 16.4  | 15.6  | 13.1  | 12.9  | 15.0  | 13.4  | 14.2  | 15.8  | 16.3  | 17.8  | 17.5  |
| (15) Precipitation                                                     | PrecAvg     | 23.7        | 1.9   | 1.4   | 1.5   | 1.3   | 2.0   | 2.4   | 2.7   | 2.6   | 2.2   | 1.9   | 1.9   | 2.0   |
|                                                                        | PrecMax     | 30.0        | 3.6   | 2.2   | 3.3   | 4.1   | 4.5   | 5.7   | 9.7   | 6.6   | 5.0   | 3.2   | 4.4   | 4.6   |
|                                                                        | PrecMin     | 17.8        | 0.2   | 0.2   | 0.4   | 0.2   | 0.6   | 0.0   | 0.0   | 0.9   | 0.6   | 0.6   | 0.2   | 0.5   |
|                                                                        | PrecStd     | 3.3         | 0.9   | 0.5   | 0.8   | 0.8   | 0.9   | 1.1   | 1.9   | 1.3   | 1.1   | 0.7   | 0.8   | 0.9   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 49.6  | 47.4  | 51.2  | 59.8  | 67.5  | 73.3  | 79.2  | 76.3  | 71.5  | 63.6  | 54.2  | 50.2  |
|                                                                        |             | MCWB        | 46.5  | 44.8  | 46.0  | 52.9  | 60.5  | 64.9  | 70.0  | 67.4  | 64.2  | 60.1  | 50.8  | 47.3  |
|                                                                        | 2%          | DB          | 45.9  | 44.8  | 47.1  | 55.3  | 64.3  | 69.4  | 76.3  | 73.8  | 68.4  | 61.2  | 51.4  | 46.6  |
|                                                                        |             | MCWB        | 43.4  | 42.7  | 44.0  | 50.2  | 58.6  | 63.2  | 68.3  | 66.3  | 62.3  | 57.9  | 49.4  | 44.4  |
|                                                                        | 5%          | DB          | 44.1  | 42.7  | 45.4  | 52.7  | 61.7  | 67.5  | 74.1  | 72.2  | 66.8  | 59.5  | 50.0  | 44.5  |
|                                                                        |             | MCWB        | 42.2  | 40.8  | 42.8  | 48.6  | 56.8  | 61.9  | 67.1  | 65.6  | 61.5  | 56.3  | 48.0  | 42.8  |
|                                                                        | 10%         | DB          | 42.1  | 40.6  | 43.9  | 50.5  | 59.8  | 65.8  | 72.0  | 70.7  | 65.3  | 57.9  | 48.7  | 43.0  |
|                                                                        |             | MCWB        | 40.5  | 39.0  | 41.6  | 47.0  | 55.5  | 60.6  | 65.6  | 65.1  | 60.6  | 54.9  | 46.9  | 41.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 46.9  | 45.2  | 46.8  | 53.3  | 62.1  | 66.3  | 71.3  | 69.2  | 65.6  | 61.3  | 51.3  | 47.4  |
|                                                                        |             | MCDB        | 49.6  | 46.7  | 49.4  | 58.6  | 65.8  | 70.9  | 77.6  | 73.4  | 69.8  | 62.9  | 52.9  | 49.0  |
|                                                                        | 2%          | WB          | 43.9  | 43.0  | 44.6  | 50.9  | 59.6  | 64.2  | 69.6  | 68.1  | 63.8  | 58.3  | 49.8  | 44.7  |
|                                                                        |             | MCDB        | 45.5  | 44.5  | 46.4  | 54.8  | 63.1  | 68.0  | 74.7  | 71.7  | 67.3  | 60.6  | 51.4  | 46.3  |
|                                                                        | 5%          | WB          | 42.2  | 41.0  | 43.0  | 49.1  | 57.8  | 62.9  | 68.1  | 67.0  | 62.6  | 56.6  | 48.6  | 43.0  |
|                                                                        |             | MCDB        | 43.8  | 42.5  | 45.1  | 52.1  | 60.9  | 66.9  | 73.5  | 70.7  | 66.0  | 59.0  | 49.9  | 44.2  |
|                                                                        | 10%         | WB          | 40.5  | 39.2  | 41.8  | 47.4  | 55.8  | 61.3  | 66.4  | 66.0  | 61.3  | 55.2  | 47.1  | 41.7  |
|                                                                        |             | MCDB        | 41.9  | 40.7  | 43.6  | 50.1  | 59.3  | 65.4  | 70.9  | 69.8  | 64.9  | 57.8  | 48.5  | 42.8  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.6   | 4.4   | 5.7   | 6.8   | 7.4   | 7.3   | 7.4   | 6.7   | 6.2   | 5.4   | 4.4   | 4.6   |
|                                                                        |             | MCDBR       | 5.5   | 5.7   | 8.3   | 9.7   | 10.4  | 9.4   | 9.4   | 8.2   | 7.9   | 6.6   | 4.7   | 4.8   |
|                                                                        | 5% WB       | MCWBR       | 4.9   | 5.1   | 6.1   | 6.0   | 6.8   | 5.9   | 5.6   | 5.4   | 5.3   | 4.9   | 4.5   | 4.2   |
|                                                                        |             | MCWBR       | 5.4   | 5.4   | 8.1   | 9.0   | 9.6   | 8.8   | 8.6   | 7.5   | 7.1   | 6.3   | 4.5   | 4.7   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.319 | 0.342 | 0.371 | 0.402 | 0.399 | 0.395 | 0.416 | 0.415 | 0.381 | 0.364 | 0.342 | 0.313 |
|                                                                        |             | taud        | 2.329 | 2.331 | 2.285 | 2.270 | 2.337 | 2.378 | 2.332 | 2.343 | 2.414 | 2.405 | 2.365 | 2.320 |
|                                                                        | Ebn at Noon |             | 199   | 235   | 253   | 261   | 269   | 271   | 262   | 255   | 249   | 224   | 188   | 175   |
|                                                                        |             | Edh at Noon | 18    | 25    | 32    | 37    | 37    | 36    | 37    | 35    | 29    | 23    | 18    | 15    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 158   | 358   | 760   | 1324  | 1693  | 1805  | 1662  | 1364  | 982   | 503   | 204   | 110   |
|                                                                        | RadStd      |             | 19    | 59    | 70    | 157   | 188   | 131   | 198   | 104   | 94    | 71    | 26    | 11    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (50 neighbors) |       | +0.74   | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | -281  | +104  | N/A   |

Nomenclature: See separate page